Cluster-Based Personalized Mental Wellness Intervention Recommendation System for Students Using Explainable Machine Learning

A mini-project synopsis report submitted by

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PROJECT TITLE

Cluster-Based Personalized Mental Wellness Intervention Recommendation System for Students Using Explainable Machine Learning

PROJECT OVERVIEW

Mental health issues such as stress, anxiety, depression, and burnout are becoming increasingly common among students due to academic pressure, financial concerns, and lifestyle changes. These challenges can affect academic performance and overall well-being. Existing support systems often provide generalized solutions that may not address individual student needs.

This project proposes a Cluster-Based Personalized Mental Wellness Intervention Recommendation System using Machine Learning and Explainable Artificial Intelligence (XAI). The system will analyze student mental health data, identify wellness profiles using clustering techniques, and recommend personalized interventions such as stress management, counseling support, sleep improvement strategies, and mindfulness activities. Explainable AI techniques will be used to provide transparent and understandable recommendations, supporting early intervention and improved student well-being.

NEED OF THE PROJECT

Student mental health concerns are increasing, while existing support systems often follow a one-size-fits-all approach. In addition, many AI-based systems lack transparency, making their recommendations difficult to understand.

The proposed system addresses these issues by identifying different student wellness profiles and providing personalized, explainable intervention recommendations. This can help educational institutions offer more effective and proactive mental health support.

OBJECTIVES

- To collect and preprocess student mental health data.
- To identify student wellness profiles using clustering techniques.
- To generate personalized intervention recommendations.

- To integrate Explainable AI for transparency.
- To evaluate the performance of the proposed system.

SCOPE OF THE PROJECT

The project focuses on analyzing student mental health data, including stress, anxiety, depression, sleep quality, academic pressure, and lifestyle factors. Machine learning techniques will be used to identify wellness profiles and recommend suitable interventions. The system serves as a decision-support tool and does not provide medical diagnosis or treatment.

EXPECTED OUTCOME

The project will develop a web-based system that identifies student wellness profiles and provides personalized mental wellness recommendations. Explainable AI will improve transparency by showing the factors influencing each recommendation.

BENEFICIARIES

- Students
- Educational Institutions
- Academic Counselors
- Mental Health Professionals
- University Wellness Centers

Student Declaration

We hereby declare that the proposed project work is original and will be carried out under the guidance of the faculty mentor.

Student Signature(s): ___________________

Date: ___________________

Faculty Guide Signature: ___________________